

GENERAL SPECIFICATION G5.35

Low Voltage Motor Control Centres

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G5.35 LOW VOLTAGE MOTOR CONTROL CENTRES

G5.35.1 General

(a) Introduction

- (i) Perform all work; design, supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, national and international codes, and as listed in General Specification G5.2 Standards, Codes and Regulations.

(b) References

- (i) The following is a list of standards which may be referenced in this Section:

1) International Electro-technical Commission (IEC)

Standard	Subject
IEC 60044	Instrument transformers.
IEC 60051	Direct Acting Indicating Analogue Electrical Measuring Instruments and Their Accessories.
IEC 60228	Conductors of insulated cables.
IEC 60238	Edison screw lamp holders.
IEC 60255	Electrical Relays.
IEC 60364	Low Voltage Electrical Installations.
IEC 60417	Graphical Symbols for Diagrams.
IEC 60439	Low Voltage Switchgear and Control gear Assemblies.
IEC 60502	Power cables with extruded insulation and their accessories for rated voltages from 1 kV (Um = 1,2 kV) up to 30 kV (Um = 36 kV) – Part 1: Cables for rated voltages of 1 kV (Um = 1,2 kV) and 3 kV (Um = 3,6 kV).
IEC 60617	Graphical Symbols for Use on Equipment.
IEC 60947	Low Voltage Switchgear and Control gear
IEC 60529	Degrees of Protection Provided by Enclosures (IP code).
IEC 60715	Dimensions of Low-Voltage Switchgear and Control gear. Standardised Mounting on Rails for Mechanical Support of Electrical Devices in Switchgear and Control gear Installations.
IEC 60838	Miscellaneous Lamp holders.
IEC 61032	Protection of Persons and Equipment by Enclosure – Probes for Verification.

Standard	Subject
IEC 61439	Low-voltage switchgear and control gear assemblies
IEC 61558	Safety of power transformers, power supplies, reactors and similar products.
IEC/TR 61641	Enclosed low-voltage switchgear and control gear assemblies - Guide for testing under conditions of arcing due to internal fault.
IEC 61643	Surge Protective Devices Connected to Low-Voltage Power Distribution Systems.

2) SPRING Singapore

Standard	Subject
CP 5	Code of Practice for Electrical Installations.
SS 551	Code of Practice for Earthing.
SS 555	Code of practice for protection against lightning
SS 538	The Maintenance of Electrical Switchgear for Voltage Up to and Including 22 kV.
CP 83	Code of Practice for Construction Computer Aided Design (CAD).
SS 97	Residual current operated circuit-breaker without integral overcurrent protection for household and similar uses (RCCBs).
SS 322	Earthing and Bonding Clamps.

- (ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..
- (iii) Wherever the requirements of the Specifications or Drawings exceed those of the above items, the requirements of the Specifications or Drawings govern. Compliance with the above standards codes and regulations is mandatory.

(c) Submittals

- (i) Shop Drawings, and Technical Data: As a minimum, submit the following items:
 - 1) Full Type Test Certificate;
 - 2) IAC (Internal Arc Containment) Test Certificate;
 - 3) Itemised bill of material;
 - 4) Descriptive information;
 - 5) Dimensional Drawings;
 - 6) Conduit and cable entrance locations;
 - 7) Bus data;

- 8) Protective Devices: Copies of time-current characteristics;
- 9) Anchoring instructions and details;
- 10) Control Diagrams:
 - Provide for all controllers;
 - In accordance with applicable requirements of IEC or BS standards;
 - Presentation in accordance with applicable requirements of SS CP 83, IEC 60417 and IEC 60617;
 - Provide blank spaces on interconnection diagrams to add control conductor code designations during installation of equipment; and
 - In addition to standard control diagrams, provide the following:
 - Remote control devices;
 - Remote indication and/or pilot lights;
 - Interconnections and interlocking circuits between starter and remote equipment;
 - Remote sensors; and
 - Tag numbers associated with all control devices and equipment.
- 11) Single-line diagrams;
- 12) Schematic (elementary) diagrams;
- 13) PLC interface schematic and diagrams and/or communication network schematics;
- 14) Manufacturer's installation instructions;
- 15) Typed Tabulation:
 - Motor name; tag (equipment) numbers as shown on the Drawings;
 - Motor kilowatt rating;
 - Nameplate full load current;
 - Measured load current and voltage;
 - Heater catalogue number;
 - Protective device trip settings; and
 - Parameters setting for MCCBs, ACBs, protection relays, energy meters, VFDs, etc.
- 16) Attach above typed, tabulated data to a copy of starter manufacturer's overload heater selection tables for the starters provided;
- 17) Operation and Maintenance Manual;
- 18) Safe operating procedures; and
- 19) Factory test reports, certified.

(d) Packing and Delivery

- (i) Delivery Equipment Sections: Established by Contractor to facilitate ingress of equipment to final installation location within the building.
- (e) Spares
 - (i) Provide spare parts, materials and special tools as listed in the Schedule of Spares, Special Tools and Consumable Items as specified in Specification P2.

G5.35.2 Products

- (a) Manufacturers
 - (i) Qualification
 - 1) Normally engaged in the manufacture of motor control equipment;
 - 2) Minimum of 10 years' experience in the design and manufacture of motor control equipment;
 - 3) Certified under ISO 9001 for manufacture of electrical apparatus; and
 - 4) Local service facility.
- (b) Motor Control Centres (MCC)
 - (i) General
 - 1) Compliant with IEC 61439 and IEC 60947;
 - 2) Operating Conditions:
 - Ambient Temperature: Maximum 40 degrees C; and
 - Equipment to be fully rated without any derating for operating conditions listed above.
 - 3) MCCs shall be intelligent Type with all motor control and monitoring occurring via a high-availability field communications network;
 - 4) Fully Type Tested design in accordance with IEC 61439 standards;
 - 5) IAC (Internal Arc Containment) Tested design in accordance with IEC 61641;
 - 6) The product along with components shall not be obsolete or slated to be obsolete;
 - 7) IEC 61439-1 Form 4b Type 6 separation requirements shall be met;
 - 8) Arc Fault Containment Design: The MCCs shall be designed and dimensioned to prevent arc fault from occurring. The MCCs shall be designed to limit the arc in the compartment where it occurs, so as to minimize the consequences in the event of arc occurrence. The MCCs shall be tested in accordance with IEC 61641 for personal safety (Criteria 1 to 5) and plant & equipment safety (Criteria 6 to 7);
 - 9) Voltage Rating: 400 V;

- 10) Short-Circuit Rating: In general, rated short-time withstand current of 65,000 amperes rms symmetrical for 3 seconds for the entire motor control centre as a complete assembly unless otherwise shown on the single line diagram. Refer to single line diagram for the rated short circuit current for the MCC, particularly for this Contract. Short circuit current withstand rating of MCC shall be verified and confirmed by the Contractor's LEW, based on the short circuit study;
 - 11) All controllers, circuit breakers, wire connections, and other devices to be front mounted and accessible unless otherwise noted;
 - 12) The clearance of active components from the switchroom floor shall be at least being 300 mm. The busbar compartment shall be at least 900 mm above switchroom floor; and
 - 13) Minimum clearance of 1500 mm in front and 600 mm at the rear sides shall be provided.
- (ii) Enclosure
- 1) Degree of Protection: IP 43 (minimum);
 - 2) Vertical Section Dimensions: Nominal 2,300 mm high, nominal 600 mm deep. Width to be standard of manufacturer but total width of any MCC shall not exceed the space shown on the Drawings without approval of the S.O.. Contractor shall coordinate the floor openings based on selected equipment;
 - 3) Construction
 - MCC cubicles shall be made of sheet steel of not less than thickness for panel front plates and 2.0 mm for panel side and rear plates and inner cubicle partitions;
 - A blank compartment with hinged door at least 300 mm high shall be provided at the base of every panel;
 - Butt sections flush, end-to-end against similar section without bolts, nuts, or cover plates causing interference;
 - Removable top cover plates and bottom cover plates; and
 - Removable plates on end panels for future bus extension.
 - 4) Section Mounting: Removable formed-steel channel sills and lifting angles;
 - 5) Horizontal Wiring Compartments: Accessible from front, full width, top and bottom;
 - 6) Vertical Wiring Compartment: Full height, isolated from unit starters with separate door;
 - 7) Unit Compartment: Individual compartments separated by steel barriers for each starter, feeder, or other unit. Separate compartment(s) for any communications or networking hardware;
 - 8) Compartment Doors: Separate hinged doors or pull-out drawers for each starter, feeder, or other unit;

- 9) Interlocking: Interlock starter and feeder doors and drawers mechanically so doors cannot be opened or drawers removed with unit energised. Provide defeater mechanism to allow intentional access at any time;
 - 10) External disconnect handles, pad lockable in OFF position; and
 - 11) Anchor Bolts: Type 316 stainless steel, sized by equipment manufacturer.
- (iii) Bus
- 1) Power Bus: Bus Bracing: 65,000 minimum amperes rms symmetrical;
 - 2) Horizontal Power Bus:
 - Three-phase, 4-wire + Earth tin-plated, copper, entire width of control centre, rated as shown on the Drawings;
 - Bus Arrangement: Neutral-B-B-G or B-B-G-Neutral, left-to-right, top-to-bottom, and front-to-rear, as viewed from front;
 - Construct to allow future extension of additional sections without any need for adapter panels;
 - Pressure type solderless lugs for each incoming line cable; and
 - Isolated from top horizontal wire way.
 - 3) Vertical Power Bus:
 - Three-phase tin-plated, copper, full height of section, rated 600 amperes, minimum; and
 - Bus bars enclosed in a partitioned compartment.
 - 4) Neutral Bus:
 - 100 percent neutral; and
 - Copper, tin-plated.
 - 5) Earth Bus: Copper, tin-plated 50 percent minimum of phase bus ampacity, entire width of control centre; and
 - 6) Phase identification shall be shown with coloured plastic sleeves. Plastic tapes are not acceptable.
- (iv) Incoming Line Section
- 1) Incoming Service Feeder: Cable entering at bottom;
 - 2) Minimum short-circuit rating of 65,000 amperes;
 - 3) ACB or MCCB circuit breaker as shown in SLD;
 - 4) Incoming Air Circuit Breakers and Bus Coupler Breakers shall be interlocked as shown on the drawings;
 - 5) ACB or MCCB shall be capable of withstanding rated fault current without damage;
 - 6) Metering as shown on single-line diagrams;
 - 7) Provide Energy Meter at each incomer of MCC as specified below:

- Phase monitoring relay to provide contact closure when source voltage is within acceptable limits; and
 - Clustered LED 3-phase indicating lamps (white, white, white) connected ahead of circuit breaker.
- (v) Feeder Unit Protective Device
- 1) Construction: As specified in paragraph Motor Controller Unit.
 - 2) Moulded Case Circuit Breaker:
 - Compliant with IEC 60947-2;
 - Thermal-magnetic trip and interrupting capacity required for connection to system with short-circuit capacity indicated; and
 - Indicate tripping by operating-handle position.
- (vi) Current Transformers and Instruments
- 1) CT Accuracy Class 0.5 for metering and 5P20 for protection, conforming to IEC 60044.
 - 2) Voltmeter:
 - Switchboard class with 240-degree minimum scale; and
 - Selector Switch: Three-phase OFF, four-position.
 - 3) Ammeter:
 - Switchboard class with 90-degree minimum scale;
 - Selector Switch: Three-phase OFF, four-position;
 - Equip metre with demand indicator; and
 - CTs with ratio indicated or as required for motor application. CTs shall meet or exceed applicable requirements of IEC 60044.
- (vii) Protection Relay for Incomer breakers (Type PR-5)
- 1) All Protection Relays shall be from the same manufacturer as switch gear unless approved by S.O.;
 - 2) External protection relays shall also be provided for all incoming and TIE feeders;
 - 3) Protection relays as minimum shall have following protections: 50, 50N, 51, 51N, 64 and 27; and
 - 4) Relay contacts shall be capable of making and breaking the maximum current which can flow in the respective circuit they control. Contacts for remote alarms and indications shall be volt free.
- (viii) The status of all incoming, TIE, and feeder circuit breakers shall be connected to the PMS. Provide all necessary i/o units and cabling works for the complete system, including daisy-chain Ethernet cabling and media converters for optical connections from MCC to PMS. PMS i/o units shall be rail-mounted remote i/o units and may be by the circuit breaker manufacturer or a reputable third party.

- (ix) PMS Communications System Interface: Each power meter, protection relay, and PMS i/o unit shall be capable of communicating to the Power Monitoring System (PMS) over an Ethernet network:
 - 1) Media Connection: Copper (internally) and Fibre optic (externally);
 - 2) Protocol: ModBus TCP/IP or other protocol to suit Power Monitoring System. Refer to Particular Specifications P6 ICA;
 - 3) Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational;
 - 4) Data Registers: As a minimum, provide the following information as data which can be queried by the host:
 - All items listed for display; and
 - All alarms.
 - 5) All power measurement and power quality monitoring data shall be sent to host computer / SCADA / PMS / EMS system as per Particular Specifications and Drawings; and
 - 6) Provide front panel accessible port connection for local interrogation and programming using a portable PC (by others) and a Windows-based program specific to this unit. Program to enable full access to unit's data and provide, as a minimum, the following features:
 - 1) View actual values;
 - 2) View unit status;
 - 3) View/edit settings (online and offline);
 - 4) View event record (for troubleshooting);
 - 5) Help screens and step-by-step tutorial; and
 - 6) Provision for installation of firmware upgrade(s) to unit.
- (x) Pushbuttons, Indicating Lights, and Selector Switches: As specified in Specification G5.10 – Basic Electrical Materials and Methods.
- (xi) Nameplates
 - 1) Laminated plastic; white, engraved to black core;
 - 2) Provide for each motor control centre and each unit;
 - 3) Engrave with inscription shown on single-line diagram;
 - 4) Provide blank nameplates on spaces for future units; and
 - 5) Attach with stainless steel pinhead screws on face of control centre.
- (xii) Motor Starter Unit
 - 1) Provide indicated individual components and control devices including pushbuttons, selector switches, indicating lights, control relays, time delay relays, and elapsed time metres as specified in General Specification G5.10 Basic Electrical Materials and Methods.

- 2) All starters shall be of the automatic type complete with shrouded flush mounting 'START/STOP' pushbuttons, Emergency Stop key pushbuttons, control circuit MCBs, auxiliary contacts, indication lamps, LCS/MCC/OFF/SCADA key selector switches as appropriate.
- 3) Star-Delta Starters:
 - Electrically and mechanically interlocked triple pole contactors, shall be arranged to automatically closed transition star-delta motor starting with an adjustable automatic timing device for switching over star to delta connection;
 - One ammeter to read phase current with matching current transformer; and
 - All other controls as required.
- 4) Construction:
 - Withdrawable Type;
 - Readily interchangeable with motor controllers of similar size;
 - Pull-apart unit control wiring terminal boards on all units;
 - Units shall be designed so they may be entirely removed from the motor control centre and replaced while the bus bars remain live without the use of tools;
 - Units shall remain solidly earthed during the removal and replacement process to ensure personnel safety;
 - IEC design, compliant with IEC 60947-4;
 - Control Transformer: Two winding, 24 V AC secondary, primary voltage as required for each starter unit with fuse on primary and MCB on secondary side with 20% spare capacity shall be provided;
 - Contactor Rating: AC3 duty, kW rated at 400 V, 5 million operations under AC3 utilisation;
 - Disconnect Type: Circuit breaker;
 - MCCBs, used with motor starter shall have adjustable magnetic trip setting adequate to protect the motor starter as well as the motor against short circuit fault;
 - Motor Protection:
 - Thermal Overload Protection for motors up to and including 55kW;
 - Multi-function protection relay for motors larger than 55 kW or where required by the drawings;
 - Manual reset overload relays shall be provided with reset pushbutton on the front of the starter unit; and

- Relay contacts shall be capable of making and breaking the maximum current which can flow in the respective circuit they control. All relay connections shall be shown on the panel wiring diagrams.
- All motor starters shall have the following indicating lights mounted on the front of the motor starter enclosure in additions to others specifically shown on control diagrams:
 - Motor Running (ON): Colour Green;
 - Motor Off: Colour Red; and
 - Fail due to overload: Colour Amber.
- 5) Variable Frequency Drive Unit (installed in MCCs):
 - Meet or exceed all applicable requirements of General Specification G5.40 LV Variable Frequency Drive Units;
 - Manufactured by the MCC manufacturer;
 - Designed for installation in motor control centre construction;
 - Disconnect Type: Circuit breaker;
 - Padlockable operating handle when de-energised;
 - Unit door interlocked to prevent opening when disconnect is in closed position; and
 - Mechanical interlocked to prevent placing disconnect in ON position when unit door is open.
- 6) Reduced Voltage Solid State Starter (Installed in MCCs):
 - Disconnect Type: Circuit breaker;
 - Rating: kW rated at 600 V;
 - Continuous Rating: 125 percent;
 - Overload Rating: 500 percent for 60 seconds;
 - Protective Features:
 - Phase loss;
 - Shorted load;
 - Shorted SCR;
 - Over temperature;
 - Overcurrent;
 - Motor overload; and
 - Earth fault protection:
 - Three-phase, line voltage limiting;

- Dual Ramp Soft-Start and Soft-Stop Controllers: Controller shall include metal oxide varistors as transient surge protection of SCR pairs rated to clamp transient voltage to 10 percent below the rated blocking voltage of the SCRs;
 - Bypass Contactor: Bypass contactor shall energise when motor is at full speed. Motor shall operate through the bypass contactor. De-energise the solid state starter until motor is again called to run; and
 - Isolation contactor as shown on the Drawings.
- Status Indicators:
 - Control power ON indication;
 - MOTOR RUN indication;
 - AT SPEED indication;
 - SHORTED SCR indication;
 - PHASE LOSS indication;
 - OVERTEMPERATURE indication;
 - OVERLOAD indication; and
 - FAULT indication.
- Front of Panel Adjustments:
 - Start voltage;
 - Ramp time;
 - Current limit;
 - Deceleration time; and
 - Stop voltage.
- Unit door interlocked to prevent opening when disconnect is in closed position;
- Mechanical interlocked to prevent placing disconnect in ON position when unit door is open;
- Minimum Dimensions: 100 mm high by full section width, less vertical wire way;
- Padlockable in OPEN position; and
- Each starter 15 kW and upwards shall be equipped with an ammeter and three-phase selector switch. Provide suitably rated current transformers with primary rating equal to 1.25 times the motor full load current rating.

- 7) Circuit Breaker:
 - Tested and co-ordinated with motor controller per IEC 60947-4-1;
 - Moulded case with manufacturer's recommended trip setting for maximum motor protection;
 - Type 2 co-ordination for all process equipment motor starters;
 - Magnetic-only type for all ACMV equipment motor starters;
 - Tripping indicated by operating-handle position;
 - Interrupting capacity required for connection to system with short-circuit capacity indicated; and
 - Monitoring contacts for open, closed, tripped connected to motor protection relay.
 - 8) Motor Thermal Protector Interface: Manual-reset interposing relay for connection to motor-mounted thermal protector system.
 - 9) Any motor control unit / starter feeding ACMV loads shall be controlled by the Contractors Building Management System and shall also provide data to the Power Monitoring System.
- (xiii) Communication with PLC
- 1) For stand alone VFD's:
 - Provide a communication Interface for each VFD. Refer to P6- Particular Specification for ICA for more details;
 - The contractor shall include all communications cabling works for standalone VFD's; and
 - See General Specification G5.40 Low Voltage Variable Frequency Drives for complete VFD requirements.
 - 2) For motor-starters within MCC's, including integral VFD's:
 - Provide Data Concentrators within MCC's to combine data interfaces of multiple starters into a single interface for each group of motors, when required to suit the capabilities of the plant control system architecture;
 - Wiring internal to MCC: Connect MCC devices to Data Concentrators within the MCC; Connect Data Concentrators to field control network Ethernet Switch in RIO panel. If the MCC internal network is a daisy-chain, it shall be arranged so that the withdrawal of one of more motor starters only interrupts communications to the withdrawn units;
 - Data Concentrator (Fieldbus Gateway) shall be as follows:
 - Data concentrators shall be intelligent devices which actively concentrate communications to reduce the communications overhead and burden on upstream PLC's;

- Factory installed within a compartment of the MCC which is dedicated to communications hardware, powered from UPS-derived supplies; and
 - Typically a product of the MCC manufacturer.
 - As a minimum, each motor starter unit shall accept the following commands from plant control system via any Data Concentrator:
 - Run Command;
 - Other commands as shown by the LCS Typical on P&IDs. These may include, but are not limited to START, STOP, TRIP, FORWARD and REVERSE; and
 - Other commands as required to fully implement the functional requirements as described in the relevant Specifications and drawings.
 - As a minimum, each motor starter unit shall provide the following status information to the plant control system via any Data Concentrator:
 - REMOTE, ON and ALARM (FAIL) status;
 - Other status as shown by the LCS Typical on P&IDs. These may include, but are not limited to FORWARD, REVERSE, HI-TORQUE, and motor current;
 - Motor power; and
 - Other status information as required to fully implement the functional requirements as described in the relevant Specifications and drawings.
- 3) All terminations shown on control diagrams for connections to a PLC, DDC, LCS or an LCP shall be factory wired to master terminal blocks located in an isolated section of the MCC.
- (xiv) Control Circuit Requirements
- 1) Disconnecting Device: Capable of de-energising external source control circuits in unit.
 - 2) Control Wiring:
 - Minimum wire size, 2.5 mm² copper;
 - Permanent sleeve type markers with wire numbers applied to each end of wires;
 - Terminate wires using insulated locking fork or ring type crimp terminals; and
 - Terminate current transformer leads on shorting type terminal blocks.
- (xv) Energy Meter- At Main Incomers (Type EM-2)
- 1) Energy Meters shall be from the same manufacturer as switch gear unless approved by S.O.;

- 2) Solid state device with easy to read displays;
- 3) Direct voltage input up to 600 V AC;
- 4) Current input via current transformer with 5 amperes secondary. CT shall be of suitable ratio with accuracy class 0.5;
- 5) Programmable current and potential transformer ratios;
- 6) Programmable limits to activate up to four alarms;
- 7) Selectable Voltage Measurements: Line-to-line or line-to-neutral and wye or delta;
- 8) Simultaneous Display:
 - V, three-phase;
 - Amperes, three-phase;
 - Kilowatts, Reactive KVA, and KVA;
 - Kilowatt-hours;
 - Power factor;
 - Frequency; and
 - Earth leakage mA.
- 9) Power Quality Monitoring:
 - Individual and total harmonics up to 31st harmonic (minimum).
- 10) Simultaneous capture of events on all channels.
 - Up to 60-cycles (minimum) each; and
 - Resolution adjustable up to 64 samples per cycles, minimum.
- 11) Sag/Swell Monitoring:
 - Record magnitude, duration and date stamp.
- 12) Communications System Interface: Each power meter shall be capable of communicating to a host computer system over an Ethernet network.
 - Media Connection: Copper (internally) and Fibre optic (externally);
 - Protocol: ModBus TCP/IP or other protocol to suit Power Monitoring System. Refer to Particular Specifications P6 ICA;
 - Network Addressing: Co-ordinated through S.O., entered and verified after units are in place and operational; and
 - Data Registers: As a minimum, provide the following information as data which can be queried by the ModBus host:
 - All items listed for display; and
 - All alarms.

- (xvi) Provide front panel accessible port connection for local interrogation and programming using a portable PC (by others) and a Windows-based program specific to this unit. Program to enable full access to unit's data and provide, as a minimum, the following features:
 - 1) View actual values;
 - 2) View unit status;
 - 3) View/edit settings (online and offline);
 - 4) View event record (for troubleshooting);
 - 5) Help screens and step-by-step tutorial;
 - 6) Provision for installation of firmware upgrade(s) to unit; and
 - 7) Software without need for licenses.
- (c) Inspection and Testing
 - (i) General: The Contractor shall provide testing and inspection check list in accordance with industrial Standards and best practices of Manufacturers.
 - 1) Test all MCC, including all components in accordance with IEC 61439; and
 - 2) Testing to be in accordance with established ISO 9001 procedures.
 - (ii) Factory Inspections
 - 1) Inspect for required construction, electrical connection, and intended function. All Source Quality Control and Factory Acceptance Tests (FAT) shall be witnessed by S.O. / Employer;
 - 2) Notify S.O. at least 2 weeks prior to any testing to allow S.O. to witness tests;
 - 3) Completely assemble MCC at the factory, and perform tests at manufacturer's shop in accordance with IEC 61439;
 - 4) Carry out routine test as part of FAT (Factory Acceptance Test) in accordance with IEC Routine Tests on all MCC;
 - 5) Conformance tests to demonstrate conformance of the equipment to applicable standards; and
 - 6) Production tests to check the quality and uniformity of the workmanship and materials used in the manufacturer of the equipment.

G5.35.3 Execution

- (a) Installation
 - (i) Install equipment in accordance with Shop Drawings and manufacturer's instructions and recommendations.
 - (ii) MCCs shall sit on concrete base of not less than 75 mm height on all four sides.
 - (iii) Secure equipment to mounting pads with anchor bolts of sufficient size and number adequate for specified seismic conditions.
 - (iv) Install equipment plumb and in longitudinal alignment with walls of room where installed.

- (v) Retighten current-carrying bolted connections and enclosure support framing and panels to manufacturer's recommendations.
- (vi) Install appropriate fire-stop materials for sealing the cable entries at MCCs to avoid any moisture ingress to the MCCs from cable chambers.
- (vii) MCCs shall have metallic handles with appropriate materials (Corrosion resistant, with adequate strength and size) at the front and back of the cubicle doors of the MCCs to facilitate the opening of the panels.
- (b) Circuit Breakers
 - (i) Field adjust trip settings of motor starter magnetic-trip-only circuit breakers.
 - (ii) Adjust to manufacturer's recommendation for full load current of motor installed.
 - (iii) Determine motor rated current from motor nameplate following installation.
- (c) Overload Relay
 - (i) Select and install overload relay heaters and adjust solid state electronic overload relays after the actual nameplate full-load current rating of motor has been determined.
 - (ii) Disable thermal overload trips for fire pump motors, smoke extraction fans etc. where required to meet FSSD requirements and to the S.O.'s approval.
- (d) Site Testing
 - (i) In accordance with General Specification G5.51 Electrical Testing.
- (e) Manufacturer's Services
 - (i) Inspection, Commissioning and Site Adjustment: In addition to the requirements of General Specification G2.25 Manufacturers' Services. An authorised service representative or training instructor of the manufacturer shall visit Site as required for each motor control centre to perform the following services:
 - 1) Verify proper installation of the equipment;
 - 2) Inspection, checking and adjusting the equipment; and
 - 3) Commissioning and field testing of the MCC, VFD controllers.
 - (ii) Training of Employer's Personnel: The authorised service representative or training instructor shall train the Employer's personnel in all aspects of drive operation and maintenance, including step-by-step troubleshooting procedures with necessary test equipment and in accordance with General Specification G2.27 Training, Advisory and Technical Support.
 - (iii) Provide Manufacturer's Certificate of Proper Installation.

END OF SECTION